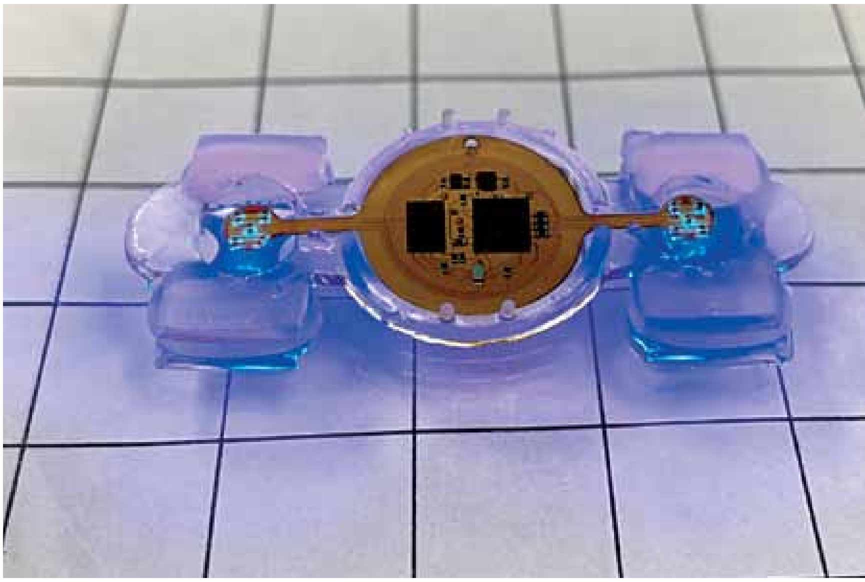
Small biological robots made with living muscle and microelectronics

Researchers at the University of Illinois Urbana-Champaign, Northwestern University and collaborating institutions have developed hybrid "eBiobots" that combine soft material, living muscle, and microelectronics. These small biological robots are powered by mouse muscle tissue grown on a soft 3D-printed polymer skeleton. To enable practical applications of the device, researchers eliminated the use of bulky batteries and tethering wires. The eBiobots use a receiver coil to harvest power and provide a regulated output voltage to power the micro-LEDs. A wireless signal to the eBiobots prompts the LEDs to pulse. The LEDs stimulate the light-sensitive engineered muscle to contract, moving the polymer legs so that the machines "walk." The micro-LEDs are so targeted that they can activate specific portions of muscle, making the eBiobot turn in a desired direction.



Remotely controlled miniature biological robots have many potential applications in medicine, sensing, and environmental monitoring (Credit: https://techxplore.com)

Shape-shifting antenna array takes shapes like curves, saddles, and spheres

Kaushik Sengupta (Credit: Princeton University)

Researchers led by Kaushik Sengupta at Princeton University have developed a new type of shape-shifting array, designed like a folded paper box called a waterbomb, which allows engineers to create a reconfigurable and adaptable radar imaging surface. The system was built by installing a new class of broadband metasurface antennas onto standard, flat panels. They connected a number of the antenna panels into a precisely designed origami surface with an offset checkerboard pattern. Through a proper sequence of folding and unfolding the panels, the array assumes a variety of different shapes like curves, saddles, and spheres. With this ability to shift and expand, the origami system offers a wider resolution and has the ability to capture complex three-dimensional scenes beyond the capability of a standard antenna array.